Global period constraints for the Q3 minimizing problem

Marked negative lattices, cyclic bubbles, and the Godeaux blowup

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Abstract

We keep the smooth-attainment problem for the trace-free Ricci functional on one blowup of the classical Godeaux surface. A marked negative lattice whose self-dual periods vanish forces the pullback canonical class to have a larger self-dual square. Combining this fact with the classical monopole curvature inequality gives a global energy restriction on a minimizing sequence, without an energy identity or a monopole extension across the limiting orbifold. In particular, two distinct degree-one (-3) classes of the classical Godeaux configuration cannot both enter such a degeneration below $8\pi^2$. A single such class requires total energy at least $64\pi^2/9$, including when the mass-replacement eigenvalue is zero. A further corollary restricts canonically marked negative cyclic chains. The results do not establish smooth attainment: canonical-orthogonal bubbles, a single permitted charged chain, and other degeneration channels remain.

1 Fixed problem and the classical input

Let X be the classical Godeaux surface and choose the blowup point outside its fifteen degree-one rational curves. Put

$$M = X\#\overline{\mathbb{CP}}^2, \qquad K = \pi^* c_1(K_X), \qquad E^2 = -1, \quad K\cdot E = 0, \quad K^2 = 1.$$

All cohomology calculations are in $H^2(M;\mathbb{R})$, so torsion is omitted. For a metric g let

$$P(g) = \int_M |\operatorname{Ric}_g^\circ|^2 dV_g, \qquad p_*(M) = \inf_{g\in \mathrm{Met}(M)} P(g), \qquad T = 2\chi(M) + 3\tau(M) = 0.$$

The supplied Q3 construction gives the strict upper bound

$$\frac{16\pi^2}{3} \le p_*(M) < 8\pi^2. \tag{1}$$

The strict inequality is an input from the general negative-Kähler–Einstein insertion construction in the supplied October 8 manuscripts. It is not reproved here. All conformal classes on this M are Yamabe-negative; a smooth P-minimizer would have constant negative scalar curvature and, if Bach-flat, would be Einstein. The positive lower bound excludes this last alternative. Thus smooth attainment on this fixed M would answer Q3. These are the same reductions used in the supplied mass-variation note [4, 5].

We use tensor norms for Ricci curvature and the Hilbert–Schmidt curvature-operator norm for Weyl curvature. The classical estimate for an actual monopole class a is

$$\frac{1}{4\pi^2}\int_M \left(\frac{s^2}{24} + 2|W^+|^2\right) dV_g \ge \frac{2}{3}\|a_g^+\|^2.$$

Together with Chern–Weil this gives

$$P(g) \ge \frac{16\pi^2}{3}\|a_g^+\|^2 - 8\pi^2 T. \tag{2}$$

The usual blowup supply selects an actual sign class with self-dual square at least $\|K_g^+\|^2$; this is the projection argument in LeBrun's Proposition 3.2, not just its final numerical bound [1]. In our case the classes are $K \pm E$, up to an overall sign. Consequently the only geometric input needed for the main argument is

$$\boxed{P(g) \geq \frac{16\pi^2}{3}\|K_g^+\|^2.} \tag{3}$$

This keeps the actual-supply and, where relevant, chamber conventions of that classical argument. It does not require FD, the O10 endpoint theorem, or any new equation on the limiting orbifold.

2 A marked lattice changes the global projection budget

Let $C_1, \ldots, C_r \in H^2(M;\mathbb{R})$ be linearly independent marked classes with negative-definite matrix

$$Q = (C_i \cdot C_j), \qquad d_i = K \cdot C_i.$$

Define

$$v = Q^{-1}d, \qquad L = \sum_i v_i C_i, \qquad \beta_Q(K) = -d^{\mathsf{T}}Q^{-1}d \geq 0, \qquad K_\perp = K - L. \tag{4}$$

These are intersection-form projections, not metric-dependent Hodge projections. Directly,

$$K_\perp \cdot C_i = 0, \qquad L^2 = -\beta_Q(K), \qquad K_\perp^2 = 1 + \beta_Q(K). \tag{5}$$

Theorem 2.1 (Finite-metric period bound)**.** *For every smooth metric on the fixed M,*

$$\boxed{P(g) \geq \frac{16\pi^2}{3}\left(\sqrt{1+\beta_Q(K)} - \|L_g^+\|\right)_+^2.} \tag{6}$$

In particular, if g_j is any sequence for which $\|(C_i)_{g_j}^+\| \to 0$ for every i, then

$$\boxed{\liminf_j P(g_j) \geq \frac{16\pi^2}{3}\left(1+\beta_Q(K)\right).} \tag{7}$$

Proof. Hodge decomposition gives

$$\|(K_\perp)_g^+\|^2 = K_\perp^2 + \|(K_\perp)_g^-\|^2 \geq 1 + \beta_Q(K).$$

Since $K^+ = (K_\perp)^+ + L^+$, the triangle inequality yields

$$\|K_g^+\| \geq \left(\sqrt{1+\beta_Q(K)} - \|L_g^+\|\right)_+.$$

Use (3). For the second conclusion, $\|L_{g_j}^+\| \leq \sum_i |v_i| \|(C_i)_{g_j}^+\| \to 0$. □

Corollary 2.2 (Restriction on a minimizing degeneration)**.** *If the periods in Theorem 2.1 vanish along a minimizing sequence, then*

$$\boxed{\beta_Q(K) \leq \frac{3p_*(M)}{16\pi^2} - 1 < \frac{1}{2}.} \tag{8}$$

Equivalently, any marked negative lattice with $\beta_Q(K) \geq 1/2$ is excluded from this kind of degeneration.

Proof. Combine (7) with (1). □

The estimate is on the original closed manifold. In particular, it is not obtained by adding a global lower bound to the energy of a bubble: that would in general count some curvature twice. No energy equality, compactness of the exterior, or existence of a bulk Einstein metric occurs in its proof.

2.1 The variant retaining the complete sign pair

When both sign classes are available for the metric under consideration, the same argument keeps possible intersections with the new exceptional class. Set $d_E = (E \cdot C_i)_i$ and, for $|t| \le 1$, let

$$B_Q(t) = 1 - t^2 - (d + td_E)^{\mathsf{T}} Q^{-1}(d + td_E).$$

Since $K + tE$ is a convex combination of $K + E$ and $K - E$,

$$\|(K + tE)^+\|^2 \le \max\{\|(K + E)^+\|^2, \|(K - E)^+\|^2\}.$$

Projecting $K + tE$ off the negative lattice as above gives, when its marked self-dual periods vanish,

$$\liminf_j P(g_j) \ge \frac{16\pi^2}{3} \max_{-1 \le t \le 1} B_Q(t). \tag{9}$$

Here the maximum is of an explicit quadratic polynomial on a compact interval. If $d_E = 0$, its maximum is $1 + \beta_Q(K)$, so it reduces to (7). The displayed sign-pair hypothesis should be retained when (9) is applied outside the present example.

3 From a genuine ALE region to vanishing self-dual periods

We now give a sufficient geometric hypothesis for the periods above. It concerns an identified marked ALE limit; it does not assert that every minimizing sequence has such a limit.

Lemma 3.1 (Compactly supported almost anti-self-dual representatives)**.** *Let (N, h) be a smooth oriented complete ALE four-manifold with negative-definite compact intersection form. Every compactly supported cohomology class has representatives ω_R, closed and supported in a compact truncation of the end, with*

$$\|\omega_R^+\|_{L^2(h)} \longrightarrow 0.$$

Proof. On an ALE four-manifold, $H^2_c(N;\mathbb{R}) \to H^2(N;\mathbb{R})$ is an isomorphism, and its classes have unique L^2 harmonic representatives [2, Lemma 1.7]. The middle-dimensional L^2 pairing agrees with the compact-support intersection form. A nonzero self-dual harmonic form would give a positive square, so negative definiteness implies that all these harmonic representatives Ω are anti-self-dual.

On an annulus $A_R = \{R < r < 2R\}$, $H^2(A_R;\mathbb{R}) = 0$ because the link is a spherical space form. Choose α_R with $d\alpha_R = \Omega$. The fixed-annulus Hodge estimate, after scaling and using the ALE convergence of the metric, permits the choice

$$\|\alpha_R\|_{L^2(A_R)} \le CR\|\Omega\|_{L^2(A_R)}.$$

One may enlarge the annulus slightly for the support of a cutoff. Let χ_R be zero near its inner boundary and one near its outer boundary, with $|d\chi_R| \le C/R$. Put $\omega_R = \Omega$ inside, $\omega_R = \Omega - d(\chi_R\alpha_R)$ on the annulus, and zero outside. This is a smooth compactly supported closed representative of the same class. Its self-dual part is confined to the annulus and satisfies

$$\|\omega_R^+\|_2 \le C\|\Omega\|_{L^2(A_R)} \longrightarrow 0.$$

The last limit follows from the finite L^2 norm. This also shows why no uniform curvature bound on the exterior of a bubble is needed. □

Proposition 3.2 (Marked ALE concentration implies the required periods)**.** *Suppose that a fixed finite collection of marked compact classes in M is carried by smooth negative-definite*

ALE limits, with smooth rescaled convergence on every fixed compact truncation and with the compact-support classes identified with the specified C_i. Then

$$\|(C_i)^+_{g_j}\| \longrightarrow 0. \tag{10}$$

The assertion applies in particular to a genuine marked smooth $\mathcal{O}(-3)$ ALE bubble. It requires neither an energy equality nor regularity of the entire limiting bulk.

Proof. Fix R and pull ω_R from the lemma into M through the marked convergence map; extend by zero. The result is a closed representative of C_i. On a closed manifold the norm of the harmonic self-dual projection is at most the self-dual norm of any closed representative. Moreover, the L^2 norm of a two-form and the Hodge star on two-forms are unchanged by constant metric scaling in dimension four. Consequently,

$$\limsup_j \|(C_i)^+_{g_j}\| \le \|\omega_R^+\|_{L^2(h)}.$$

Now let $R \to \infty$. □

For a tree with orbifold intermediate levels, the relevant markings and period limits must still be obtained on the original manifold. The proposition covers smooth ALE pieces or any other region for which (10) has been established; it does not turn an arbitrary orbifold or collapsed limit into such a piece.

4 Consequences for the classical Godeaux configuration

4.1 One or two (-3) classes

A degree-one rational curve C in X has

$$C^2 = -3, \qquad K \cdot C = 1, \qquad E \cdot C = 0.$$

For its rank-one lattice, $Q = [-3]$ and $\beta_Q(K) = 1/3$. Hence:

Corollary 4.1. *If this marked class enters a degeneration covered by* (10)*, then*

$$\liminf_j P(g_j) \ge \frac{64\pi^2}{9} = \frac{8}{9}(8\pi^2). \tag{11}$$

This includes the first-order zero-mode case of the mass-replacement argument.

For two disjoint such classes the charge is $2/3$, so instead

$$\liminf_j P(g_j) \ge \frac{80\pi^2}{9} = \frac{10}{9}(8\pi^2) > 8\pi^2. \tag{12}$$

This contradicts the inherited strict upper bound for a minimizing sequence. Although two separate scalar-flat Kähler $\mathcal{O}(-3)$ fillings have combined local P-energy only $16\pi^2/3$, their simultaneous marked degeneration is globally excluded by (12).

4.2 At most one of the fifteen original degree-one lines

The classical Godeaux surface has fifteen degree-one rational curves L_i^m, with $m \in \{-1, 0, 1\}$ and $i \in \mathbb{Z}/5$. Their intersections are computed in [3, Proposition 4.3]. Distinct curves have intersection $s \in \{0, 1, 2, 5\}$; each has square -3 and canonical degree one. If $s < 3$, the pair matrix is negative definite and

$$\beta_Q(K) = \frac{2}{3 - s}.$$

Thus the lower bounds, normalized by $8\pi^2$, are

s	0	1	2
$\beta_Q(K)$	2/3	1	2
$\liminf P/(8\pi^2)$	10/9	4/3	2

and are all incompatible with a minimizing sequence. If $s = 5$, then $(C + D)^2 = 4$, so $\|(C+D)^+\| \geq 2$ for every metric; the two self-dual periods cannot both tend to zero even without an energy estimate.

Corollary 4.2. *Along a minimizing sequence on this marked Godeaux blowup, at most one of the fifteen original degree-one rational-curve classes can have vanishing self-dual period. In particular at most one can enter the negative-definite ALE degenerations described in the previous section.*

The statement refers to these fixed classes, not to a classification of all embedded spheres or all possible markings on M.

5 Canonically marked cyclic chains

For a proposed cyclic plumbing let

$$Q_{ii} = -n_i, \quad Q_{i,i+1} = Q_{i+1,i} = 1, \quad n_i \geq 2, \qquad K \cdot C_i = n_i - 2, \qquad E \cdot C_i = 0.$$

The pairings are part of the marking hypothesis. They hold by adjunction for a rational curve configuration inherited from the unblown surface; they must not be assigned to an arbitrary smooth bubble without proof. Assume these periods tend to zero along a minimizing sequence.

Theorem 5.1 (Restriction on cyclic chain data)**.** *Every connected chain with a nonzero canonical pairing must have the form*

$$[-3, \underbrace{-2, \ldots, -2}_{r-1}]$$

or its reversal. For this chain,

$$\beta_Q(K) = \frac{r}{2r+1}, \qquad \liminf_j \frac{P(g_j)}{8\pi^2} \geq 1 - \frac{1}{6r+3}. \tag{13}$$

At most one mutually orthogonal component can have a nonzero canonical pairing. Chains of (-2) *classes are not excluded by this theorem.*

Proof. Write $A = -Q$, so A has diagonal n_i and off-diagonal -1, and set $d_i = n_i - 2 \geq 0$. Its charge has the variational expression

$$d^{\mathsf{T}} A^{-1} d = \max_z \{2d^{\mathsf{T}} z - z^{\mathsf{T}} A z\}.$$

Testing with $z_i = d_i/n_i$ gives

$$\beta_Q(K) \geq \sum_i \frac{(n_i - 2)^2}{n_i}.$$

If some $n_i \geq 4$, the right side is at least one. If two weights are at least three, it is at least 2/3. Both contradict (8). Hence there is exactly one weight three and all other weights are two.

Let the two (-2) tails have lengths l and r'. The endpoint diagonal entry of the inverse A_l Cartan matrix is $l/(l+1)$, as follows directly from the tridiagonal recurrence. Its Schur complement gives

$$\beta_Q(K) = \frac{1}{3 - l/(l+1) - r'/(r'+1)}.$$

If both tails are nonempty, this is at least $1/2$, again excluded. The three must therefore be an end vertex. With total length r the charge is $r/(2r+1)$, and Theorem 2.1 gives (13). For orthogonal components the charges add. Each nonzero component has charge at least $1/3$, so two would violate (8). □

For lengths $r = 1, 2, 3, 4$, the normalized thresholds are respectively $8/9, 14/15, 20/21, 26/27$. A verified upper bound $p_*/(8\pi^2) \le U < 1$ restricts any such surviving chain to

$$6r + 3 \le \frac{1}{1-U}.$$

This is a restriction on specified marked cyclic data, not an assertion that all minimizing bubbles are cyclic or canonically marked.

6 Relation to the mass zero mode and to attainment

The supplied mass-variation result uses a different input: a smooth P-critical orbifold point, marked matching, and either a global energy identity or the stated local lower comparison. It gives

$$P(g_{\lambda,t}) = P(g_{\mathrm{orb}}) + \frac{8\pi^2}{3} - 24\pi^2 m(g_t)\mu\lambda^2 + o(\lambda^2),$$

where positive and negative masses are available on the same filling. Under those hypotheses, a minimizing degeneration requires $\mu = 0$ [5]. The present period argument neither changes this expansion nor uses its coefficient. It imposes an independent condition on the same marked degeneration:

$$\boxed{\mu = 0, \qquad p_*(M) \ge 64\pi^2/9}$$

when both sets of hypotheses apply. Two of the original (-3) classes cannot occur together, regardless of their local values of μ.

The single surviving charged case is not excluded because $64\pi^2/9 < 8\pi^2$. If a future strict upper bound below $64\pi^2/9$ is proved on this same M, all rank-one canonical-degree-one (-3) degenerations satisfying the period hypothesis will then be excluded. No such upper bound is proved here.

If K pairs trivially with a negative lattice, its charge in (4) is zero. Thus the method does not exclude canonical-orthogonal ADE lattices or bubbles with no real second cohomology. It also does not establish marked convergence for every minimizing sequence or rule out partial collapse. The separate zero-mode calculation found in the companion research records likewise retains an undetermined full fourth-order gluing coefficient [6]; a negative inner coefficient alone cannot remove this final case.

The current fixed task is unchanged: obtain smooth global attainment on M. The increment proved here is a constraint on which actual marked negative-lattice degenerations a minimizing sequence could have. It uses the original closed-manifold monopole inequalities throughout, rather than presupposing an extension of those inequalities to an unknown orbifold bulk.

References

[1] C. LeBrun, *Ricci Curvature, Minimal Volumes, and Seiberg–Witten Theory*, Invent. Math. **145** (2001), 279–316. Propositions 3.1–3.2 and Corollary 3.4. `https://arxiv.org/abs/math/0003068`.

[2] H.-J. Hein and C. LeBrun, *Mass in Kähler Geometry*, Comm. Math. Phys. **347** (2016), 183–221. Lemma 1.7. `https://arxiv.org/abs/1507.08885`.

[3] C. Böhning, H.-C. Graf von Bothmer and P. Sosna, *On the derived category of the classical Godeaux surface*, Adv. Math. **243** (2013), 203–231. Section 4, Proposition 4.3. `https://arxiv.org/abs/1206.1830`.

[4] Supplied project manuscripts, *Four Public Problems: Overnight Results, Proof Mechanisms, Remaining Connections, and Questions for Further Research*, and *Mathematical Source Companion*, 8 October 2026. The strict upper construction and the Q3 variational reduction are inputs.

[5] Supplied continuation, *Q3: Mass Variation and a Cyclic Degeneration Obstruction*, 8 October 2026, Sections 2–5.

[6] Related continuation, *Q3: The Zero Mode and the Proposed Horikawa Candidate*, 8 October 2026, Sections 4–6.